

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

(Level/CO) Marks

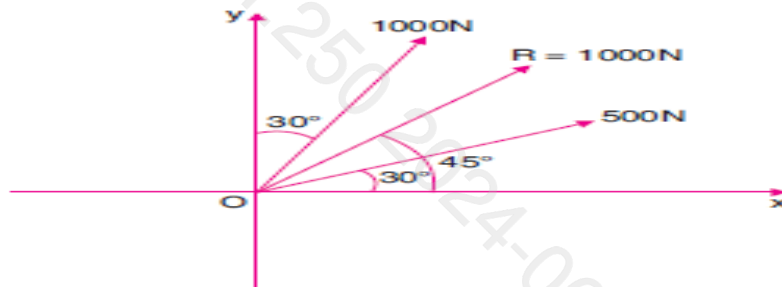
Q. 1 Solve Any Two of the following.

12

- A) Two forces acting on a body are 500 N and 1000 N as shown in Figure below. Determine the third force F such that the resultant of all the three forces is 1000 N directed at 45° to x-axis.

CO1

6



- B) Find the magnitude of the two forces, such that if they act at right angles, their resultant is $\sqrt{10}$ N. But if they act at 60° , their resultant is $\sqrt{13}$ N.

CO1

6

- C) a) What is the difference between coplanar concurrent forces and non concurrent forces?
b) Give an example of Free Body Diagram for a beam subjected to UDL and point load. Assume suitable support conditions.

CO1

3+3=6

Q.2 Solve Any Two of the following.

12

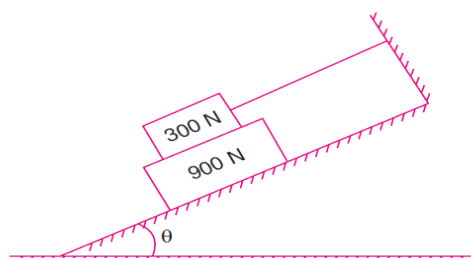
- A) Explain Angle of Friction and Angle of Repose with diagram.
B) What should be the value of θ in Figure below that will make the motion of 900 N block down the plane to impend? The coefficient of friction for all contact surfaces is $1/3$.

CO2

6

CO2

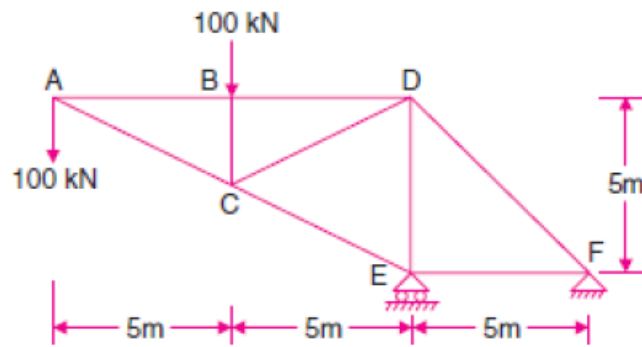
6



C) Find any 6 member force of following truss. Assume suitable data if required.

CO2

6



Q. 3 Solve Any Two of the following.

12

A) A small steel ball is shot vertically upwards from the top of a building 25 m above the ground with an initial velocity of 18 m/sec.

CO3

3+3=

6

(a) In what time, it will reach the maximum height?

(b) How high above the building will the ball rise?

B) A wheel rotating about a fixed axis at 20 revolutions per minute is uniformly accelerated for 70 seconds during which it makes 50 revolutions. Find the (i) angular velocity at the end of this interval and (ii) time required for the velocity to reach 100 revolutions per minute.

CO3

6

C) Two ships move from a port at the same time. Ship A has velocity of 30 kmph and is moving in N 30° W while ship B is moving in south-west direction with a velocity of 40 kmph. Determine the relative velocity of A with respect to B and the distance between them after half an hour.

CO3

6

Q.4 Solve Any Two of the following.

12

A) Explain Mass moment of inertia and D'Alembert's principle with suitable example.

CO4

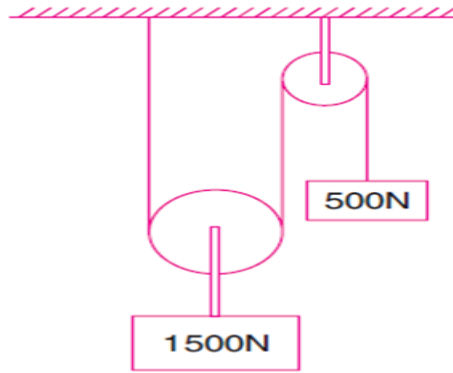
3+3=

6

B) Determine the tension in the string and accelerations of blocks A and B weighing 1500 N and 500 N connected by an inextensible string as shown in Figure below Assume pulleys as frictionless and weightless.

CO4

6



- C) A flywheel weighing 50 kN and having radius of gyration 1m loses its speed from 400 rpm to 280 rpm in 2 minutes. Calculate
- CO4 2+2+
2=6
- (i) the retarding torque acting on it
(ii) change in its kinetic energy during the above period
(iii) change in its angular momentum during the same period.

Q. 5 Solve Any Two of the following.

12

- A) A block weighing 2500 N rests on a level horizontal plane for which coefficient of friction is 0.20. This block is pulled by a force of 1000 N acting at an angle of 30° to the horizontal. Find the velocity of the block after it moves 30 m starting from rest. If the force of 1000 N is then removed, how much further will it move? Use work energy method.
- CO5 6
- B) A ball of mass 2 kg, moving with a velocity of 3 m/s, impinges on a ball of mass 4 kg moving with a velocity of 1 m/s. The velocities of the two balls are parallel and inclined at 30° to the line of joining their centres at the instant of impact. If the coefficient of restitution is 0.5, Explain
- CO5 3+3=6
- I. Direction, in which the 4 kg ball will move after impact;
II. Velocity of the 4 kg ball after impact;
- C) State and prove work energy principle.
- CO5 6

***** End *****